DATA STRUCTURES LAB SE COMP

Group B Practical no 4

```
from collections import deque


# Create queue
call_queue = deque()


def addCall():
    customerID = input("Enter Customer ID: ")
    callTime = int(input("Enter Call Time (minutes): "))


    call_queue.append((customerID, callTime))
    print("Call added successfully.")


def answerCall():
    if not call_queue:
        print("Queue is empty. No call to answer.")
    else:
        customerID, callTime = call_queue.popleft()
        print("Answered Call")
        print("Customer ID:", customerID)
        print("Call Time:", callTime, "minutes")
```

```
def viewQueue():
    if not call_queue:
        print("Queue is empty.")
    else:
        print("\nCurrent Call Queue:")
        for customerID, callTime in call_queue:
            print("Customer ID:", customerID,
                  "| Call Time:", callTime, "minutes")


def isQueueEmpty():
    if not call_queue:
        print("Queue is empty.")
    else:
        print("Queue is not empty.")



# Menu
while True:
    print("\n----- CALL CENTER QUEUE -----")
    print("1. Add Call")
    print("2. Answer Call")
    print("3. View Queue")
    print("4. Check Queue Empty")
    print("5. Exit")
```

```
    choice = int(input("Enter your choice: "))

    if choice == 1:
        addCall()

    elif choice == 2:
        answerCall()

    elif choice == 3:
        viewQueue()

    elif choice == 4:
        isQueueEmpty()

    elif choice == 5:
        print("Program terminated.")
        break

    else:
        print("Invalid choice.")
```

Output

----- CALL CENTER QUEUE -----

1. Add Call

2. Answer Call

3. View Queue

4. Check Queue Empty

5. Exit

Enter your choice: 1

Enter Customer ID: C101

Enter Call Time (minutes): 5

Call added successfully.

Enter your choice: 1

Enter Customer ID: C102

Enter Call Time (minutes): 8

Call added successfully.

Enter your choice: 3

Current Call Queue:

Customer ID: C101 | Call Time: 5 minutes

Customer ID: C102 | Call Time: 8 minutes

Enter your choice: 2

Answered Call

Customer ID: C101

Call Time: 5 minutes